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Impact of transverse and longitudinal dispersion on first-order degradation rate constant estimation

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Abstract

A two-dimensional analytical model is employed for estimating the first-order degradation rate constant of hydrophobic organic compounds (HOCs) in contaminated groundwater under steady-state conditions. The model may utilize all aqueous concentration data collected downgradient of a source area, but does not require that any data be collected along the plume centerline. Using a least squares fit of the model to aqueous concentrations measured in monitoring wells, degradation rate constants were estimated at a former manufactured gas plant (FMGP) site in the Midwest U.S. The estimated degradation rate constants are 0.0014, 0.0034, 0.0031, 0.0019, and 0.0053 day^{-1} for acenaphthene, naphthalene, benzene, ethylbenzene, and toluene, respectively. These estimated rate constants were as low as one-half those estimated with the one-dimensional (centerline) approach of Buscheck and Alcantar [Buscheck, T.E., Alcantar, C.M., 1995. Regression techniques and analytical solutions to demonstrate intrinsic bioremediation. In: Hinchee, R.E., Wilson, J.T., Downey, D.C. (Eds.), Intrinsic Bioremediation, Battelle Press, Columbus, OH, pp. 109–116] which does not account for transverse dispersivity. Varying the transverse and longitudinal dispersivity values over one order of magnitude for toluene data obtained from the FMGP site resulted in nearly a threefold variation in the estimated degradation rate constant—highlighting the importance of reliable estimates of the dispersion coefficients for obtaining reasonable estimates of the degradation rate constants. These results have significant implications for decision making and site management where overestimation of a degradation rate may result in remediation times and bioconversion factors that exceed expectations. For a complex source area or non-steady-state plume, a superposition of

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analytical models that incorporate longitudinal and transverse dispersion and time may be used at sites where the centerline method would not be applicable.
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1. Introduction

When natural attenuation is considered as a remedial alternative or final remedial step in the restoration of contaminated properties, the estimation of degradation rates may play an important role in decision making and site management. Buscheck and Alcantar (1995) developed an equation that requires only several contaminant concentration measurements along the centerline of a dissolved plume to estimate a first-order degradation rate constant based on a one-dimensional transport equation under the assumption that the contaminant plume is at steady-state. The Buscheck and Alcantar method has been used or reported by a variety of authors including Chapelle (1999), Azadpour-Keeley et al. (2001), Zamfirescu and Grathwohl (2001), Suarez and Rifai (2002), Wilson and Kolhatkar (2002), and Bockelmann et al. (2003). The Buscheck and Alcantar method, however, does not account for transverse dispersion (perpendicular to the flow direction) and does not utilize data collected away from the plume centerline. McNab and Dooher (1998) showed that the Buscheck and Alcantar (1995) method is easily subject to misinterpretation because transverse dispersion and temporal effects can produce centerline concentration profiles that resemble a degrading contaminant even in the absence of degradation.

The objective of this paper is to show how a degradation rate constant may be estimated using a familiar analytical model based on work by Domenico (1987) that includes longitudinal and transverse dispersion and time. The method described herein may also be utilized at sites where a well-defined plume centerline is not apparent, possibly due to the existence of several contaminant sources. A homogeneous porous aquifer with uniform groundwater flow, groundwater velocity high enough that molecular diffusion can be ignored, reasonably well-known longitudinal and transverse dispersivities, a constant source, first-order degradation, and a reversible linear sorption isotherm are assumed. We do not consider macrodispersion in heterogeneous media (e.g., see Cirpka and Kitanidis, 2000), but assume that dispersion is reasonably well approximated by well-known approximate solutions to an analytical advection–dispersion transport equation.

The first-order degradation process is a mathematical simplification used to model the effects of a variety of biogeochemical processes which have been widely discussed (e.g., McNab and Dooher, 1998; Chapelle, 1999; Guoping et al., 1999). The degradation term, as used herein and by Domenico and Schwartz (1990), Wiedemeier et al. (1999), and Buscheck and Alcantar (1995), among others, implicitly incorporates all processes other than dispersion and reversible sorption that remove contamination from the aqueous phase, including biodegradation, loss due to volatilization, hydrolysis reactions, and uptake by plants or other organisms, insofar as these processes affect the measured concentrations that are used to estimate the degradation rate constant. Without methods to measure or estimate the contribution of these individual processes to the overall contaminant loss, it is

not possible to reliably separate them on the basis of aqueous concentrations alone. In practice, processes other than degradation are often assumed to be negligible, and the rate constant is generally considered to be a biodegradation rate constant (e.g., Zamfirescu and Grathwohl, 2001; Bockelmann et al., 2003).

2. One-dimensional reactive transport

For a linear sorption isotherm, $C^* = K_d C$, where C^* is the mass of contaminant sorbed per mass of dry soil, K_d is the distribution coefficient, and C is the aqueous contaminant concentration. Under this model, the average velocity of a contaminant front, or center of mass of a contaminant plume, in one-dimensional flow without degradation is given by

$$v_c = \frac{v_x}{R} \tag{1}$$

where $R = 1 + (B_d K_d / n)$ is the retardation factor, B_d is the dry soil bulk density, n is the effective porosity, and v_x is the average groundwater velocity (Darcy velocity divided by n) in the x direction. Throughout this paper, groundwater flow is assumed to be in the x direction. A retarded contaminant front velocity occurs because sorption removes some contamination from the aqueous phase as the contaminated groundwater moves into uncontaminated soil, resulting in a decrease in aqueous concentration and an associated apparent contaminant velocity that is less than the average groundwater velocity.

When transverse dispersion, if allowed to act a large distance perpendicular to the flow direction, and/or degradation occur, a plume generated from a constant source in steady groundwater flow will approach steady-state conditions given sufficient time. As the plume approaches steady-state, the apparent velocity of the contaminant front will decrease, eventually becoming zero at steady-state. Although a plume at steady-state does not appear to be moving, net movement of contaminant molecules dissolved in the groundwater within the plume occurs at the groundwater velocity. Accordingly, as a plume approaches or reaches a steady-state condition, the retarded contaminant velocity given by Eq. (1) has no physical meaning. Due to nonconstant external factors, it is unlikely that a plume at a field site will ever achieve a true steady-state condition. However, a quasi steady-state condition may be achieved. Measurements at individual monitoring wells can be used to assess the stability of a plume by testing for statistically significant trends with time (e.g., see Wiedemeier and Haas, 2002; Suarez and Rifai, 2002). While the steady-state assumption may be reasonable in cases where no significant trends are found, large temporal variability sometimes observed in contaminant concentration data may make the detection of significant trends difficult.

A one-dimensional advection–dispersion equation that includes sorption and degradation is given by Fetter (1993) as

$$\frac{\partial C}{\partial t} = D_x \frac{\partial^2 C}{\partial x^2} - v_x \frac{\partial C}{\partial x} - \frac{B_d}{n} \frac{\partial C^*}{\partial t} + \left(\frac{\partial C}{\partial t}\right)_{\text{rxn}} \tag{2}$$

where $(\partial C/\partial t)_{\text{rxn}}$ is the rate of change in aqueous concentration due to degradation reactions, and D_x is the dispersion coefficient in the x direction (the groundwater flow

direction). It is important to distinguish between the $\partial C/\partial t$ and the $(\partial C/\partial t)_{\mathrm{rxn}}$ terms in Eq. (2). The $\partial C/\partial t$ term describes a rate of change in concentration due to all processes acting in the system. This term is observable as repeated measurements over time at any given location and, although the precision and accuracy of sample analytical results may affect the interpretation of data, it is zero for a steady-state plume. The $(\partial C/\partial t)_{\mathrm{rxn}}$ term describes a rate of change due to chemical and/or biologically mediated reactions which may occur. In a steady-state plume, the rate of contaminant degradation is matched by the rate of contaminant influx minus outflux at each location, so that the observable concentration at each location remains constant. Accordingly, for a steady-state plume, a change in concentration due to degradation cannot be observed by measuring contaminant concentrations over time in a single monitoring well, even though degradation may be occurring.

Assuming a linear isotherm and following the work of van Genuchten and Wagenet (1989), first-order degradation terms that describe the rate of contaminant mass lost per unit volume of water for the aqueous and solid phases, denoted by subscripts "aq" and "s", respectively, can be developed as:

$$\left(\frac{\partial C}{\partial t}\right)_{\mathrm{rxn}} = \left(\frac{\partial C}{\partial t}\right)_{\mathrm{rxn,aq}} + \frac{B_{\mathrm{d}}}{n}\left(\frac{\partial C^*}{\partial t}\right)_{\mathrm{rxn,s}} = -\lambda_{\mathrm{aq}}C - \lambda_{\mathrm{s}}\frac{B_{\mathrm{d}}}{n}K_{\mathrm{d}}C = -\lambda C \tag{3}$$

where λ_{aq} $[\mathrm{T}^{-1}]$ is a rate constant for first-order degradation in the aqueous phase, λ_{s} $[\mathrm{T}^{-1}]$ is a rate constant for first-order degradation in the solid phase and

$$\lambda = \lambda_{\mathrm{aq}} + \lambda_{\mathrm{s}}\frac{B_{\mathrm{d}}K_{\mathrm{d}}}{n} \tag{4}$$

is an overall (observable) first-order degradation rate constant that describes what is observable in aqueous phase concentrations due to degradation in both the aqueous and solid phases. It accounts for degradation in the solid phase by relating the solid phase concentration to the aqueous concentration via the isotherm, as described in Eq. (3). Note that if λ_{aq} and λ_{s} are identical (e.g., with radioactive decay), Eq. (4) reduces to $\lambda = R\lambda_{\mathrm{aq}}$. However, for chemical or microbial degradation the two rate constants are likely to have different values (van Genuchten and Wagenet, 1989). Unless the relationship between λ_{aq} and λ_{s} is known, it is not possible to estimate λ_{aq} or λ_{s} separately on the basis of aqueous phase concentrations alone, but it is possible to estimate λ. Substituting Eq. (3) into Eq. (2) and assuming a linear sorption isotherm gives the well-known one-dimensional advection–dispersion equation with a first-order degradation process:

$$R\frac{\partial C}{\partial t} = D_x\frac{\partial^2 C}{\partial x^2} - v_x\frac{\partial C}{\partial x} - \lambda C \tag{5}$$

A solution to Eq. (5) is (modified after Bear, 1979)

$$C(x,t) = \left(\frac{C_0}{2}\right)\left\{\exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1-\sqrt{1+\frac{4\lambda\alpha_x}{v_x}}\right]\right\}\mathrm{erfc}\left(\frac{x - v_{\mathrm{c}}t\sqrt{1+4\lambda\alpha_x/v_x}}{2\sqrt{\alpha_x v_{\mathrm{c}} t}}\right)\right.$$
$$\left. + \exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1+\sqrt{1+\frac{4\lambda\alpha_x}{v_x}}\right]\right\}\mathrm{erfc}\left(\frac{x + v_{\mathrm{c}}t\sqrt{1+4\lambda\alpha_x/v_x}}{2\sqrt{\alpha_x v_{\mathrm{c}} t}}\right)\right\} \tag{6}$$

where x is the distance downgradient from the source, t is the time since source emplacement, erfc(·) is the complimentary error function, and α_x is the longitudinal dispersivity ($D_x = \alpha_x v_x$). The terms to the right of the plus sign in Eq. (6) are generally neglected (e.g., see Bear, 1979; Zamfirescu and Grathwohl, 2001). Letting t approach infinity in Eq. (6) gives the steady-state solution

$$C(x) = C_0 \exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1 - \sqrt{1 + \frac{4\lambda\alpha_x}{v_x}}\right]\right\} \tag{7}$$

Note that neither the retarded contaminant velocity nor the retardation factor explicitly appears in the steady-state solution given by Eq. (7). However, the distribution coefficient (K_d) is implicitly incorporated into this equation via the relationship in Eq. (4), indicating that retardation does affect the steady-state concentration profile.

If λ_{aq} is assumed to be equal to λ_s, then Eq. (5) may be rewritten as

$$\frac{\partial C}{\partial t} = \frac{D_x}{R}\frac{\partial^2 C}{\partial x^2} - \frac{v_x}{R}\frac{\partial C}{\partial x} - \lambda_{aq} C \tag{8}$$

Eq. (8) is the form of the differential equation that Buscheck and Alcantar (1995) and Wiedemeier et al. (1999) use (although Wiedemeier et al. used a three-dimensional version that incorporates transverse dispersion). Buscheck and Alcantar (1995) assumed that measured concentrations along the centerline of a real plume at steady-state would follow a first-order overall attenuation process given by

$$C(x) = C_0 \exp[-kx/v_x] \tag{9}$$

where x is the distance downgradient for groundwater moving at velocity v_x and k (T^{-1}) is a first-order attenuation rate constant that includes attenuation due to longitudinal dispersion and degradation with distance away from the contaminant source. The value of k is determined from the slope of the natural logarithm of observed concentrations down the centerline of a contaminant plume ($k = -$ (slope) $\times v_x$). Setting Eq. (7) equal to Eq. (9) and solving for λ gives

$$\lambda = \frac{v_x}{4\alpha_x}\left[\left(1 + 2\alpha_x\left(\frac{k}{v_x}\right)\right)^2 - 1\right] \tag{10}$$

Eq. (10) is a general solution for overall degradation as observed by aqueous phase concentrations. Degradation may occur in the aqueous phase, solid phase or both phases in accordance with Eq. (4). Starting with Eq. (8), whereby $\lambda_{aq} = \lambda_s$ is assumed, results in the more familiar solution given by Buscheck and Alcantar (1995) as

$$\lambda_{aq} = \frac{v_c}{4\alpha_x}\left[\left(1 + 2\alpha_x\left(\frac{k}{v_x}\right)\right)^2 - 1\right] \tag{11}$$

Eqs. (10) and (11) are consistent with $\lambda = \lambda_{aq} R$ under the assumption that $\lambda_{aq} = \lambda_s$.

Eq. (10) can be rewritten by expanding the square and simplifying, giving

$$\lambda = k\left(1 + k\left(\frac{\alpha_x}{v_x}\right)\right) \tag{12}$$

Assuming that the concentrations decrease downgradient, then k is always positive and λ in Eq. (12) will always be greater than k, indicating that the overall degradation rate constant, λ, is greater than the first-order attenuation rate constant, k. This occurs in this model because the effect of longitudinal dispersion is to spread contamination further downgradient than would advective transport alone, thus increasing downgradient concentrations and decreasing the magnitude of the slope which is used to estimate the first-order attenuation rate constant, k.

3. Multi-dimensional reactive transport

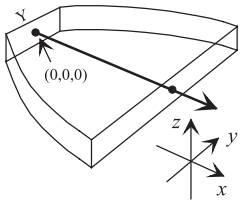
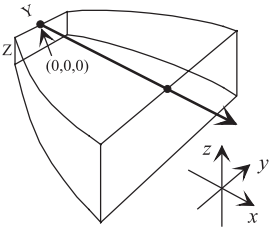
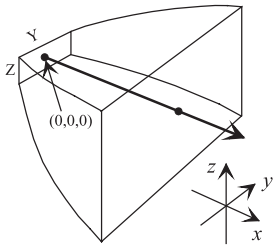
The overall degradation rate constant, λ, will, in most cases, be overestimated by Eq. (10) because it does not account for transverse dispersion, which lowers the observed concentrations along the plume centerline, resulting in a greater slope (more negative) than can be attributed to degradation alone. Also, Eq. (10) is not applicable if the plume is not at steady-state. Accordingly, flow equations that include longitudinal and transverse dispersion, sorption, degradation, and, possibly, time must be considered to better justify estimation of the degradation rate from groundwater concentrations collected at a contaminated site.

Domenico and Schwartz (1990) give an approximate solution to the multidimensional groundwater flow problem that includes sorption, longitudinal and transverse dispersion and first-order degradation, as described by the differential equation

$$R\frac{\partial C}{\partial t} = D_x\frac{\partial^2 C}{\partial x^2} + D_y\frac{\partial^2 C}{\partial y^2} + D_z\frac{\partial^2 C}{\partial z^2} - v_x\frac{\partial C}{\partial x} - \lambda C \tag{13}$$

where D_y and D_z are the dispersion coefficients in the y and z directions, respectively (see the coordinate system in Table 1). The approximate solutions given by Domenico and Schwartz (1990) assume a constant source concentration, C_0, at $x = 0$ along a plane that is perpendicular to the direction of groundwater travel (the x direction) that is Y units wide and Z units high, and that contaminant spreading can occur in the $\pm y$ directions with either no spreading in the z direction, spreading in the $-z$ direction, or spreading in the $\pm z$ directions. Approximate solutions to Eq. (13) based on Domenico and Schwartz (1990) for these dispersion scenarios are given in Table 1. Domenico and Schwartz (1990) also provided solutions for other aquifer scenarios. Note that the use of v_x and v_c in the equations in Table 1 differs from others because the λ used in this paper (i.e., Eq. (4)) is based on the assumption that λ_{aq} and λ_s are indeterminable from aqueous concentrations alone. With the assumption that $\lambda = \lambda_{aq} = \lambda_s$, as in Wiedemeier et al. (1999) and Domenico and Schwartz (1990), the corresponding equations herein and in those references yield identical results.

Table 1
Approximate solutions for multi-dimensional flow for several dispersion scenarios[a]

Dispersion scenario	Approximate solution
	Case a: Dispersion in the $\pm y$ directions only. $$C(x,y,z,t) = \left(\frac{C_0}{4}\right)\exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1-\sqrt{1+\frac{4\lambda\alpha_x}{v_x}}\right]\right\}\mathrm{erfc}\left(\frac{x-v_c t\sqrt{1+4\lambda\alpha_x/v_x}}{2\sqrt{\alpha_x v_c t}}\right) \cdot\left\{\mathrm{erf}\left(\frac{y+Y/2}{2\sqrt{\alpha_y x}}\right)-\mathrm{erf}\left(\frac{y-Y/2}{2\sqrt{\alpha_y x}}\right)\right\}$$
	Case b: Dispersion in the $\pm y$ and the $-z$ directions. $$C(x,y,z,t) = \left(\frac{C_0}{8}\right)\exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1-\sqrt{1+\frac{4\lambda\alpha_x}{v_x}}\right]\right\}\mathrm{erfc}\left(\frac{x-v_c t\sqrt{1+4\lambda\alpha_x/v_x}}{2\sqrt{\alpha_x v_c t}}\right) \cdot\left\{\mathrm{erf}\left(\frac{y+Y/2}{2\sqrt{\alpha_y x}}\right)-\mathrm{erf}\left(\frac{y-Y/2}{2\sqrt{\alpha_y x}}\right)\right\} \times\left\{\mathrm{erf}\left(\frac{z+Z}{2\sqrt{\alpha_z x}}\right)-\mathrm{erf}\left(\frac{z-Z}{2\sqrt{\alpha_z x}}\right)\right\}$$
	Case c: Dispersion in the $\pm y$ and the $\pm z$ directions. $$C(x,y,z,t) = \left(\frac{C_0}{8}\right)\exp\left\{\left(\frac{x}{2\alpha_x}\right)\left[1-\sqrt{1+\frac{4\lambda\alpha_x}{v_x}}\right]\right\}\mathrm{erfc}\left(\frac{x-v_c t\sqrt{1+4\lambda\alpha_x v_x}}{2\sqrt{\alpha_x v_c t}}\right) \cdot\left\{\mathrm{erf}\left(\frac{y+Y/2}{2\sqrt{\alpha_y x}}\right)-\mathrm{erf}\left(\frac{y-Y/2}{2\sqrt{\alpha_y x}}\right)\right\} \times\left\{\mathrm{erf}\left(\frac{z+Z/2}{2\sqrt{\alpha_z x}}\right)-\mathrm{erf}\left(\frac{z-Z/2}{2\sqrt{\alpha_z x}}\right)\right\}$$

[a] Adapted from Domenico and Schwartz (1990).

The equations in Table 1 indicate that a steady-state condition in a contaminated aquifer may be achieved without degradation because of dilution associated with dispersion, although this condition would take an infinite amount of time to achieve. Even within a finite time, lack of precision in analytical equipment and natural variation in measured contaminant concentrations may indicate that a plume has reached a quasi steady-state. Because this apparent steady-state can be achieved without degradation, it is important that other lines of evidence be developed to directly verify or indirectly indicate that degradation of the contaminants of concern are or may be occurring under the specific conditions existing at the site.

Although finite element or finite difference models may have advantages for modeling some sites for which the assumptions underlying the analytical models described herein are not well met, it is reasonable to utilize an analytical solution to model some sites. However, as Bockelmann et al. (2003) note, many sites show complex source zones with

the possibility of different contaminant plumes migrating downgradient from a source area. While a single application of one of the models in Table 1 may be insufficient to adequately model some sites, a superposition of several analytical models may be constructed to reasonably approximate the source area(s) and some types of spatially varying aquifer and groundwater flow conditions to effectively model a plume.

A least squares criterion may be utilized to find the degradation rate in an equation from Table 1, or in a superposition of two or more of these equations, that gives the best fit to data from multiple monitoring wells. Examination of these equations shows that both the degradation rate constant and the longitudinal dispersivity appear in the exponential term. It is not difficult to show that for any given longitudinal dispersivity, there is a λ that makes the exponential term argument a constant times x. Accordingly, a least squares solution cannot be used to estimate both λ and α_x simultaneously. Thus, an independent estimate of α_x is necessary, and the value assumed for α_x will affect the estimate of λ. It is possible to estimate the transverse dispersivity values, α_y and α_z, as unknown parameters using a least squares method to provide a best fit to the available monitoring well data. Alternatively, Wiedemeier et al. (1999) provide guidance in selecting the longitudinal and transverse dispersion parameters.

4. Midwest U.S. FMGP site

Data have been selected from a former manufactured gas plant (FMGP) site in the Midwest U.S. to demonstrate the methodology described here. The data were collected from monitoring wells placed downgradient and to the sides of a contaminant plume emanating from an area contaminated with coal tar in subsurface soils beneath the former gas works. The gas plant operated from about 1870 to 1937. During the years between 1951 and 1991, several petroleum service stations known to have had leaking underground storage tanks were operated on the FMGP property and an upgradient adjacent property causing a commingling of petroleum and FMGP contaminants. Fig. 1 shows a site plan with approximate source area, plume position, and monitoring locations. Source materials continue to persist in the subsurface soils at the site. Selected data from the site are shown in Table 2; these data are averages from several years of monitoring events. While some individual results vary considerably over time, no downgradient wells show statistically significant trends with time. Accordingly, the average concentrations are assumed to represent an approximate steady-state configuration. Monitoring wells MW1A, MW8A, and MW7B are downgradient of the source area near the centerline of the plume. Monitoring wells MW5A, MW6B, and MW12 are downgradient from the source area and near the edges of the plume. The (x, y) coordinates listed in Table 2 are with respect to the groundwater flow direction, as indicated in Case a of Table 1.

Following guidance from Wiedemeier et al. (1999), a longitudinal dispersivity of one-tenth the approximate plume length was assumed ($\alpha_x = 10$ m). A transverse dispersivity, α_y, equal to 1.5 m was found to provide a good overall fit to the data and is within the range of one-tenth to one-third the longitudinal dispersivity, as suggested by Wiedemeier et al. (1999). The groundwater velocity was $v_x = 0.09$ m/day. The source area width was taken to be $Y = 27$ m. A least squares criterion was used to fit a model to contaminant concen-

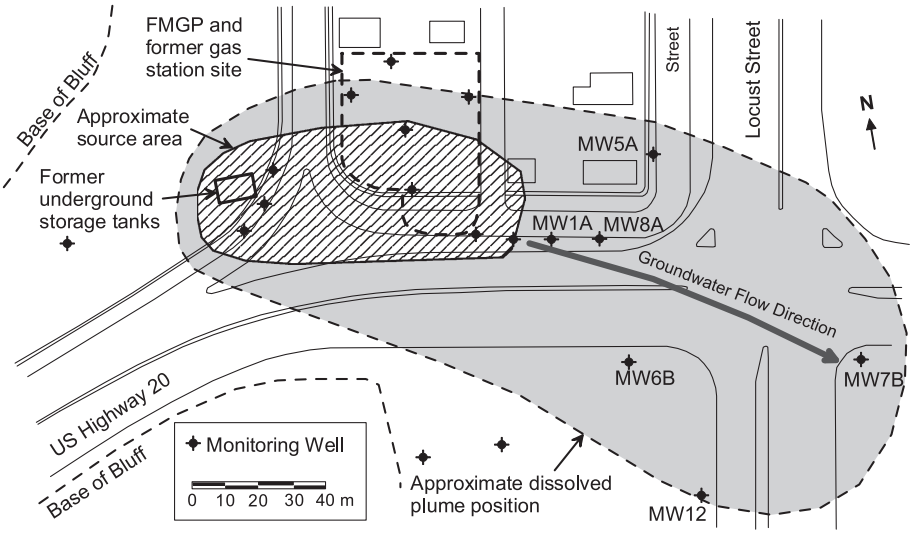


Fig. 1. Overview of the field site with the location of the source, dissolved plume, and monitoring wells.

trations using the equation in Case a of Table 1. The source concentration, C_0, and the overall degradation rate constant, λ, were estimated. The measured concentrations, modeled concentrations, and least squares parameter estimates are shown in Table 2. Computations were done using Solver in Microsoft® Excel. For comparison, the overall degradation rate constant estimated from Eq. (10) (longitudinal dispersion only) based on data from the plume centerline monitoring wells MW1A, MW8A, and MW7B are also listed in Table 2.

Table 2
Concentration data and model output from the FMGP site[a]

Sample ID	x (m)	y (m)	Acenaphthene (μg/l)		Naphthalene (μg/l)		Benzene (μg/l)		Ethylbenzene (μg/l)		Toluene (μg/l)	
			Actual	Model	Actual	Model	Actual	Model	Actual	Model	Actual	Model
MW1A	12.2	0	155	150	3338	3363	7188	7058	2173	2307	770	772
MW8A	23.2	5.2	103	110	2118	2065	4317	4453	1849	1614	420	416
MW7B	111.9	0	26	21	59	98	329	263	73	220	6.6	6.9
MW6B	47.2	−25.9	9.1	14	52	182	462	417	35	190	7.7	28
MW12	81.7	−57.0	8.6	0.2	5.7	1.2	3	2.9	3.2	1.8	1.9	0.1
MW5A	27.7	37.5	2.1	0.5	4.9	9.3	14	20	4.3	7.6	4.2	1.8
Estimation output												
C_0 (LS[b], μg/l)			183		4946		10075		2935		1312	
λ (LS[b], day^{-1})			0.0014		0.0034		0.0031		0.0019		0.0053	
λ (Eq. (10), day^{-1})			0.0018		0.0051		0.0035		0.0042		0.0063	

[a] The detection limit for PAHs was 2 μg/l; actual values near 2 μg/l may include nondetected results.
[b] Least squares estimates.

Table 3
Sensitivity of toluene degradation rate constant (day^{-1}) to dispersivity at the FMGP site

Longitudinal dispersivity α_x (m)	λ by least squares estimation: transverse (horizontal) dispersivity, α_y (m)			λ by Eq. (10) (longitudinal dispersion only)
	0.3	1.5	3	
3	0.0056	0.0042	0.0039	0.0049
10	0.0075	0.0053	0.0048	0.0063
20	0.0101	0.0068	0.0062	0.0083
30	0.0127	0.0083	0.0075	0.0103

Comparison of the degradation rates estimated by least squares versus Eq. (10) in Table 2 shows that inclusion of transverse dispersivity in the parameter estimation results in lower degradation rate constants, as expected. On the basis of these data, however, the results of Eq. (10) appear to agree relatively well (within a factor of just over 2) with the least squares estimates.

To assess the effects of the choice of dispersivity values on the degradation rate constant estimation, both longitudinal dispersivity and transverse dispersivity were varied by one order of magnitude and the degradation rate constant was estimated for toluene using both a least squares fit and Eq. (10) for each case. The results, listed in Table 3, indicate that the least squares estimated degradation rate constant varies by a factor of about three over this range of dispersivity values. Some of the least squares estimates are slightly greater than the Eq. (10) estimate shown in Table 2 because of the influence of the non-centerline data utilized in the least squares fits but not in Eq. (10).

5. Discussion and conclusions

The contaminant transport models listed in Table 1 show that given sufficient time, an aqueous phase plume will approach a steady-state configuration regardless of the presence or absence of degradation. As a dissolved plume approaches steady-state, the velocity of the plume front approaches zero. Transverse dispersion decreases the concentrations along the plume centerline below that which would be observed with longitudinal dispersion and degradation alone. In most cases, this leads to an overestimation of the first-order degradation rate constant when applying the Buscheck and Alcantar centerline method. As long as the centerline concentrations decrease away from the source, the centerline method will always give a positive degradation rate constant, even if there is no degradation occurring. In addition, if the location of the plume centerline is not well known, the existing monitoring wells are not placed near the centerline, the plume is not at steady-state, or the centerline is ill-defined because of the presence of several source areas, then the application of the Buscheck and Alcantar centerline method may produce misleading results.

The first-order degradation rate constant described herein is an overall rate constant that is estimable on the basis of observed aqueous phase concentrations. The overall rate constant incorporates degradation in both the aqueous phase and solid phase, but does not make any assumption about their individual magnitudes or relative importance to the

overall contaminant degradation. The overall degradation rate constant may be estimated by fitting a multi-dimensional reactive transport model to contaminant concentration data collected from site monitoring wells using a least squares fitting method. The method presented here utilizes data from all monitoring wells located downgradient of the contaminant source area, may incorporate more than one source area located within a site, and may be easily programmed into a computer spreadsheet using existing functions and optimization routines. At sites where the groundwater flow is relatively steady and the aquifer is relatively homogeneous, a single analytical model or a superposition of several models may be used to model contaminant plumes at sites with simple or complex source areas, and may include time to account for the temporal effects of an expanding plume. The models listed in Table 1 are candidates for such model fitting, but are themselves approximate solutions to a mathematical simplification of underlying processes, and the accuracy of output from such models will depend on how well the site conditions match the underlying model assumptions. For example, the values selected for the transverse and longitudinal dispersion will affect the degradation rate constant estimate. Varying both the transverse and longitudinal dispersivity values over one order of magnitude for toluene data from the FMGP site resulted in a threefold variation in the estimated degradation rate constant. These findings have significant implications for decision making and site management where overestimation of the degradation rate may result in remediation times and bioconversion factors that exceed expectations.

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